United States Patent Application Publication	20250284301
Kind Code	A1
Publication Date	September 11, 2025
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FAUCET DEVICE

Abstract

A faucet device according to an embodiment includes a water spout part, a hot water and cold water mixing unit, an actuator, a controller, and an operation unit. The hot water and cold water mixing unit includes a body casing, a thermo-sensitive energizing part, and a valve body. The controller is configured to: execute a water spouting mode and a calibration mode, the water spouting mode being a mode for driving, in receiving the information on the set temperature, the actuator to a predetermined position or by a predetermined drive amount that are corresponding to the set temperature to adjust a position of the valve body in an axial direction of the valve body; and the calibration mode is a mode for adjusting setting of the predetermined position or the predetermined drive amount in accordance with at least one of an arrangement environment and product unevenness.

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Family ID: **1000008486876**

Appl. No.: **19/059672**

Filed: **February 21, 2025**

Foreign Application Priority Data

JP	2024-036269	Mar. 08, 2024

Publication Classification

Int. Cl.: **G05D23/13** (20060101); **F16K11/00** (20060101); **F16K11/10** (20060101)

U.S. Cl.:

CPC **G05D23/1333** (20130101); **F16K11/10** (20130101); **F16K19/006** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-036269, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiment of the disclosure relates to a faucet device.

BACKGROUND

[0003] Conventionally, in a faucet device including an electronic temperature controlling valve that causes an actuator to drive a Shape Memory Alloy (SMA) thermo-valve, there has been known that feedback control is executed on the actuator in order to adjust a spouted water temperature in spouting water (see Japanese Laid-Open Patent Publication No. 2011-021328).

[0004] In the above-mentioned faucet device, there has been a problem that in a case where temperature control by feedback control of the actuator and temperature adjustment by the SMA are executed in an overlapped manner, it takes a long time interval until a spouted water temperature stabilizes. Moreover, there has been a problem that in a case without the feedback control; a hot water temperature, a hot water pressure, a cold water temperature, and a cold water pressure from a hot water supplying path and a cold water supplying path are different for each scene, and thus temperature adjustment is not completely executed by spontaneous operations of the SMA thermo-valve depending on conditions thereof, thereby leading to difference in a spouted water temperature in some cases.

SUMMARY

[0005] A faucet device according to one aspect of an embodiment includes: a water spout part that spouts water towards an interior of a bathroom; a hot water and cold water mixing unit that mixes hot water and cold water to be supplied to the water spout part; an actuator that causes the hot water and cold water mixing unit to operate; a controller configured to control driving of the actuator; and an operation unit configured to transmit information on a set temperature to the controller based on an operation of a user, wherein the hot water and cold water mixing unit includes: a body casing in which a hot water inlet, a cold water inlet, and a mixed hot water and cold water outlet are formed; a thermo-sensitive energizing part that changes in accordance with a temperature of mixed hot water to be capable of adjusting open degrees of the hot water inlet and the cold water inlet; and a valve body that is incorporated into the body casing to be able to slide along an axial direction of the body casing to adjust the open degrees of the hot water inlet and the cold water inlet, and the controller is further configured to: execute a water spouting mode and a calibration mode, the water spouting mode being a mode for driving, in receiving the information on the set temperature, the actuator to a predetermined position or by a predetermined drive amount that are corresponding to the set temperature to adjust a position of the valve body in an axial direction of the valve body; and the calibration mode is a mode for adjusting setting of the predetermined position or the predetermined drive amount in accordance with at least one of an arrangement environment and product unevenness.

[0006] The faucet device is capable of adjusting, in accordance with each of scenes, a hot water temperature, a hot water pressure, a cold water temperature, and a cold water pressure from the hot water supplying path and the water supplying path which are different for each scene, and individual difference in temperature adjustment of an SMA thermo-valve. Thus, the faucet device causes the actuator to move the SMA thermo-valve only when changing a set temperature, and

further executes temperature adjustment by using the SMA thermo-valve during spouting of water, so that it is possible to execute temperature adjustment having high accuracy according to the set temperature. The faucet device is capable of achieving, as much as possible, features to be capable of shortening a time interval until a temperature becomes stable in a case where temperature change and/or pressure fluctuation occurs in supplied cold water or supplied hot water during spouting of water.
[0007] The faucet device further includes: a temperature sensor arranged on upstream or downstream of the hot water and cold water mixing unit, wherein in the calibration mode, setting of the predetermined position or the predetermined drive amount is adjusted based on information from the temperature sensor.
[0008] In the faucet device, in entering the calibration mode, a predetermined position or a predetermined drive amount is automatically adjusted. Thus, the faucet device is capable of improving usability of a user.
[0009] In the calibration mode, the controller is further configured to: cause the actuator to operate while spouting water from the water spout part.
[0010] The faucet device drives the actuator while recognizing environment in actually spouting water, so that it is possible to improve the accuracy in the calibration.
[0011] The calibration mode can be executed by an operation with respect to the operation unit.
[0012] Calibration can be executed by a remote controller that is used in common water spouting operations. Thus, the faucet device does not need a dedicated setting remote controller.
[0013] The controller is further configured to: in the calibration mode; search equal to or more than three temperature changing points; and adjust the predetermined position or the predetermined drive amount based on the specified temperature changing points.
[0014] The faucet device uses equal to or more than three temperature changing points so as to improve the accuracy in calibration.
[0015] The controller is further configured to: in the calibration mode; acquire a first temperature that is a temperature at a temperature changing point where searching and specifying a temperature lower than a predetermined temperature as a target temperature; a second temperature that is a temperature at a temperature changing point where searching and specifying the predetermined temperature as a target temperature; and a third temperature that is a temperature at a temperature changing point where searching and specifying a temperature higher than the predetermined temperature as a target temperature; and adjust setting of the predetermined position or the predetermined drive amount based on the second and the first temperatures, and the second and the third temperatures.
[0016] It is true that there presents a case where characteristics of the thermo-sensitive energizing part change between under and over the predetermined temperature, the faucet device adjusts setting of a position or a drive amount that are predetermined on the basis of a temperature within a range lower than the predetermined temperature and a temperature within a range higher than the predetermined temperature, so that it is possible to prevent a set temperature from shifting in executing temperature adjustment crossing the predetermined temperature.
[0017] The controller is further configured to: in the calibration mode; search two temperature changing points; and adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing points.
[0018] The faucet device is capable of executing calibration without taking time, while keeping the accuracy in calibration near the predetermined temperature.
[0019] The controller is further configured to: in the calibration mode; search a single temperature changing point; and adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing point.
[0020] The faucet device is capable of shortening a time interval until completion of calibration.
[0021] The actuator includes a motor, and in the calibration mode; a temperature sensor arranged

on downstream of the hot water and cold water mixing unit detects a spouted water temperature, and the controller is further configured to: adjust setting of the predetermined position or the predetermined drive amount based on a drive amount of the motor and the spouted water temperature in a case where the motor takes the drive amount.
[0022] The faucet device executes simple operational processing on the basis of a drive amount of the motor and a spouted water temperature, so that it is possible to simplify software processing.
[0023] In the calibration mode; a temperature sensor arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and the controller is further configured to: determine whether or not the hot water supplying temperature is equal to or less than a predetermined temperature.
[0024] The faucet device is capable of recognizing whether or not a hot water supplying temperature is an appropriate temperature, and further is capable of informing a user of whether or not the hot water supplying temperature is an appropriate temperature in accordance with a situation.
[0025] The controller is further configured to: in the calibration mode; in a case where the hot water supplying temperature is equal to or less than the predetermined temperature, execute the water spouting mode by using a table that stores therein the set temperature, and the predetermined position or the predetermined drive amount in association with each other.
[0026] In a case where a hot water supplying temperature is equal to or lower than an appropriate temperature, the faucet device is capable of reducing a case where lukewarm hot water is spouted. Thus, the faucet device is capable of avoiding reduction in usability of a user.
[0027] The actuator includes a motor, and the controller is further configured to: during the calibration mode, not release excitation of the motor.
[0028] In a case where the excitation is released after driving the motor by a small step, there presents possibility that spring back occurs, which causes the motor to return to the last position due to load that is applied to the motor; however, it is possible to remove effects due to the above-mentioned spring back, and further to execute precise temperature controlling during execution of calibration. Thus, the faucet device is capable of improving the accuracy in calibration.
[0029] In the calibration mode; a temperature sensor that is arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and the controller is further configured to: start to adjust setting of the predetermined position or the predetermined drive amount after the hot water supplying temperature becomes stable.
[0030] The faucet device is capable of precisely executing temperature controlling in calibration while reducing effects due to heat absorption of cold cast metal, and starting up of the hot water supplying device. Thus, the faucet device is capable of improving the accuracy in calibration.

Description

BRIEF DESCRIPTION OF DRAWING(S)
[0031] FIG. **1** is a schematic diagram illustrating one example a bath unit in which a faucet device is arranged according to an embodiment;
[0032] FIG. **2** is a front view illustrating a remote controller according to the embodiment;
[0033] FIG. **3** is a block diagram illustrating the outline of a faucet device according to the embodiment;
[0034] FIG. **4** is a perspective view illustrating a faucet body;
[0035] FIG. **5** is a perspective view illustrating a combination faucet unit;
[0036] FIG. **6** is a perspective cross sectional view taken along a line VI-VI illustrated in FIG. **5**;
[0037] FIG. **7** is a flowchart illustrating a processing procedure for temperature controlling in a water spouting mode;

[0038] FIG. **8** is a flowchart illustrating a processing procedure for an initial process and a supplied hot water temperature determining process in a calibration mode;
[0039] FIG. **9** is a flowchart illustrating a processing procedure for a normal optimizing process in the calibration mode;
[0040] FIG. **10** is a flowchart illustrating a processing procedure for a temperature adjustment table assigning process in the calibration mode;
[0041] FIG. **11** is a diagram illustrating calculation of an assignment factor;
[0042] FIG. **12** is a diagram illustrating calculation of an assignment factor; and
[0043] FIG. **13** is a diagram illustrating calculation of an assignment factor.

DESCRIPTION OF EMBODIMENT(S)

[0044] As illustrated in FIG. **1**, a faucet device **1** according to an embodiment is arranged in a bath unit **10**, for example. FIG. **1** is a schematic diagram illustrating one example the bath unit **10** in which the faucet device **1** is arranged according to the embodiment.

[0045] An illustrated orthogonal coordinate system prescribes the positive direction of an X-axis as “left direction”, and further prescribes the negative direction of the X-axis as “right direction”. The orthogonal coordinate system prescribes the positive direction of a Y-direction as “back direction”, and further prescribes the negative direction of the Y-direction as “front direction”. The orthogonal coordinate system prescribes the positive direction of a Z-axis as “upward”, and further prescribe the negative direction of the Z-axis as “downward”. Therefore, in the following explanation, an X-axis direction may be referred to as a left-right direction, a Y-axis direction may be referred to as a front-back direction, and a Z-axis direction may be referred to as an up-and-down direction.

[0046] The bath unit **10** includes a bathtub **11**, a first counter **12**, a second counter **13**, and the faucet device **1**. Note that in the following, in a case where not differentiating cold water spouted from the faucet device **1**, and mixed hot water obtained by mixing hot water and cold water with each other to be spouted from the faucet device **1**, they are comprehensively referred to as “warm/cold water”.

[0047] The first counter **12** is attached to a wall part **14***a* of the bath unit **10**. The first counter **12** protrudes towards the interior of a bathroom from the wall part **14***a*. The first counter **12** is arranged above a washing place floor **15** of the bath unit **10**. The first counter **12** houses therein a faucet body **3** of the faucet device **1**. A hot water waiting water spout port **25***a* is arranged at a lower end of the first counter **12**. The hot water waiting water spout port **25***a* is arranged in the first counter **12** such that a water spouting direction of warm/cold water is downward.

[0048] The second counter **13** is attached to the wall part **14***a*. The second counter **13** protrudes towards the interior of the bathroom from the wall part **14***a*. The second counter **13** is arranged above the first counter **12**. For example, the second counter **13** extends over the bathtub **11**. Note that it is sufficient that the second counter **13** does not extend over the bathtub **11**.

[0049] The second counter **13** houses therein a part of the faucet device **1**. Specifically, the second counter **13** houses therein a part of a faucet **21** in the faucet device **1**, and a part of a hand shower **22** in the faucet device **1**. A water spout port **21***a* of the faucet **21** is arranged in the second counter **13** to be exposed therefrom. The water spout port **21***a* of the faucet **21** is arranged in the second counter **13** such that a water spouting direction of warm/cold water is downward.

[0050] A shower hose **22***a* of the hand shower **22** in the faucet device **1** is connected with the second counter **13**. The shower hose **22***a* is connected to a shower water guiding channel housed in the second counter **13**.

[0051] To a ceiling **16** of the bath unit **10**, an overhead shower **23** of the faucet device **1** and a warm pillar **24** of the faucet device **1** are attached. The overhead shower **23** and the warm pillar **24** are integrally configured.

[0052] The overhead shower **23** spouts warm/cold water towards a broad range of a user that is broader than a case of the hand shower **22**. For example, the overhead shower **23** configured such that warm/cold water falls on the whole body of a user. The warm pillar **24** collects and arranges warm/cold water into a single water flow so as to spout it therefrom. In other words, the warm pillar **24** arranges and spouts warm/cold water such that consecutive warm/cold water falls down in a columnar manner.

[0053] A remote controller **4** (namely, operation unit) of the faucet device **1** is attached to a wall part **14***b* of the bath unit **10**. The remote controller **4** may be attached to the wall part **14***a* to which the first counter **12** and the second counter **13** are attached.

[0054] The remote controller **4** receives various operations of a user with respect to the faucet body **3**. Specifically, the remote controller **4** receives a setting operation of a temperature (namely, set temperature) of warm/cold water in the faucet body **3**. The remote controller **4** receives a setting operation of a flow volume of warm/cold water in the faucet body **3**. The remote controller **4** receives a switching operation between spouting and stopping of warm/cold water. The remote controller **4** receives a switching operation of a water spouting part of warm/cold water. In a case where being operated by a user, the remote controller **4** transmits an operation signal according to a corresponding operation to a controller **7** (see FIG. **3**) of the faucet device **1**.

[0055] For example, as illustrated in FIG. **2**, the remote controller **4** includes a temperature adjusting button **41**, a water amount adjusting button **42**, and a switching button **43**. FIG. **2** is a front view illustrating the remote controller **4** according to the embodiment.

[0056] The temperature adjusting button **41** is a button for adjusting a temperature of mixed hot water in the faucet body **3**. The temperature adjusting button **41** includes a high temperature button **41***a* and a low temperature button **41***b*. The high temperature button **41***a* is a button for increasing a temperature of mixed hot water. The low temperature button **41***b* is a button for reducing a temperature of mixed hot water. Note that the remote controller **4** causes a first display **45***a* to display a setting state of a temperature of mixed hot water. In a case where the high temperature button **41***a* or the low temperature button **41***b* is operated, displaying of the first display **45***a* is changed in accordance with an operation with respect to the buttons **41***a* and **41***b*.

[0057] Note that a temperature of mixed hot water in the faucet body **3** can be adjusted within a predetermined temperature range. In a case where a set temperature of mixed hot water becomes lower than the lowest temperature of the predetermined temperature range by an operation with respect to the low temperature button **41***b*, hot water is not mixed with cold water in the faucet body **3** so as to spout the cold water.

[0058] The water amount adjusting button **42** is a button for adjusting a flow volume of warm/cold water to be spouted from the faucet device **1**. The water amount adjusting button **42** includes a water increasing button **42***a* and a water reducing button **42***b*. The water increasing button **42***a* is a button for increasing a flow volume of warm/cold water. The water reducing button **42***b* is a button for reducing a flow volume of warm/cold water. The remote controller **4** causes a second display **45***b* to display a setting state of a water amount of warm/cold water. In a case where the water increasing button **42***a* or the water reducing button **42***b* is operated, displaying of the second display **45***b* changes in accordance with operations with respect to the buttons **42***a* and **42***b*. Note that a flow volume of warm/cold water spouted from the faucet device **1** can be adjusted within a predetermined flow volume range.

[0059] The switching button **43** is a button for switching between spouting and stopping of warm/cold water in the faucet device **1**. The switching button **43** is a button for changing a water spouting part of warm/cold water in the faucet device **1**. The switching button **43** includes a faucet button **43***a*, a hand shower button **43***b*, an overhead shower button **43***c*, and a warm pillar button **43***d*. Each of the buttons **43***a* to **43***d* is switched between “ON” and “OFF” by a user depressing the corresponding button.

[0060] In a case where the buttons **43***a* to **43***d* of the switching button **43** are turned “OFF”, warm/cold water is not spouted. In other words, the faucet device **1** is in a water stopping state.

[0061] In a case where one of the switching button **43** is depressed and further the depressed switching button **43** is turned “ON” from a state where the buttons **43***a* to **43***d* of the switching

button **43** are turned “OFF”, warm/cold water is spouted. In other words, the faucet device **1** is turned from a water stopping state into a water spouting state.
[0062] In a case where one of the switching button **43** is depressed in a state where another of the switching button **43** is turned “ON”, the switching button **43** to be turned “ON” is changed so as to change a water spouting part of warm/cold water.
[0063] For example, in a case where the hand shower button **43***b* is turned “ON”, warm/cold water is spouted from the hand shower **22**. In this state, in a case where the faucet button **43***a* is depressed, the hand shower button **43***b* is turned “OFF”, and further the faucet button **43***a* is turned “ON”. Thus, a water spouting part of warm/cold water is changed from the hand shower **22** into the faucet **21**, and warm/cold water is spouted from the faucet **21**.
[0064] In a case where the switching button **43** having been turned “ON” is depressed again, the depressed switching button **43** is turned “OFF”, and the buttons **43***a* to **43***d* of the switching button **43** are turned “OFF” so as to stop spouting warm/cold water. In other words, the faucet device **1** is turned from a water spouting state into a water stopping state.
[0065] Note that each of the buttons **43***a* to **43***d* is configured such that a user is able to identify “ON” and “OFF” states of the corresponding button. For example, the switching button **43** having been turned “ON” lights up, and further the switching button **43** having been turned “OFF” blacks out.
[0066] Herein, the faucet device **1** is explained as one example which is capable of spouting warm/cold water from the overhead shower **23**, the warm pillar **24**, the hand shower **22**, and the faucet **21**; however, not limited thereto. For example, the faucet device **1** may have a configuration without the overhead shower **23** and the warm pillar **24**.
[0067] Next, the outline of the faucet device **1** according to the embodiment will be explained with reference to FIG. **3**. FIG. **3** is a block diagram illustrating the outline of the faucet device **1** according to the embodiment. In FIG. **3**, flows of warm/cold water are indicated by arrows of solid lines, and communication lines are indicated by dashed lines.
[0068] The faucet device **1** includes a plurality of water spout parts **2**, the faucet body **3**, the remote controller **4**, and a communication unit **5**.
[0069] The plurality of water spout parts **2** includes the faucet **21**, the hand shower **22**, the overhead shower **23**, the warm pillar **24**, and the hot water waiting water spout port **25***a* (see FIG. **1**) that is connected to a remaining water discharging flow path **25**.
[0070] The faucet body **3** includes a combination faucet unit **50**, a spout water switching unit **30**, and the controller **7**.
[0071] The combination faucet unit **50** includes a hot water and cold water mixing unit **80**, a flow volume adjusting unit **100**, a temperature sensor **48**, and a temperature sensor **49**. Hot water is supplied to the hot water and cold water mixing unit **80** from a hot water supplying source **37**. Cold water is supplied to the hot water and cold water mixing unit **80** from a water supply source **44**. The temperature sensor **48** is arranged on upstream of the hot water and cold water mixing unit **80** so as to detect a temperature of hot water that is supplied from the hot water supplying source **37**. The temperature sensor **49** is arranged on downstream of the hot water and cold water mixing unit **80** so as to detect a temperature of warm/cold water that is spouted from the hot water and cold water mixing unit **80**. A water stopping valve **38** is arranged on a hot water supplying path **39** between the hot water and cold water mixing unit **80** and the hot water supplying source **37**. A water stopping valve **45** is arranged on a water supplying path **46** between the hot water and cold water mixing unit **80** and the water supply source **44**.
[0072] The hot water and cold water mixing unit **80** mixes hot water supplied from the hot water supplying source **37** and cold water supplied from the water supply source **44** with each other. Specifically, the hot water and cold water mixing unit **80** switches between whether or not hot water is mixed to cold water. The hot water and cold water mixing unit **80** adjusts a ratio of hot water to be mixed to cold water, so as to adjust a temperature of mixed hot water.

[0073] The hot water and cold water mixing unit **80** includes a motor **62**. The motor **62** is a stepping motor, for example, and a rotational position (namely, drive amount) of the motor **62** is controlled by the number of steps. The motor **62** operates in accordance with an operation with respect to the temperature adjusting button **41** of the remote controller **4** so as to drive a temperature adjusting valve **82** (see valve body illustrated in FIG. **6**), and the hot water and cold water mixing unit **80** switches between whether or not hot water is mixed to cold water.
[0074] The motor **62** operates in accordance with an operation with respect to the temperature adjusting button **41** of the remote controller **4** so as to drive the temperature adjusting valve **82** (see FIG. **6**), and the hot water and cold water mixing unit **80** adjusts a ratio of hot water to be mixed to cold water. Even in a case where the temperature adjusting button **41** is not operated, for example, when a temperature of hot water changes so as to change a temperature of mixed hot water, the hot water and cold water mixing unit **80** adjusts a ratio between a flow volume of hot water and a flow volume of cold water in accordance with the temperature of mixed hot water, so that it is possible to automatically adjust a temperature of mixed hot water.
[0075] Warm/cold water flows into the flow volume adjusting unit **100** from the hot water and cold water mixing unit **80**. In a case where warm/cold water is spouted from the water spout parts **2**, the flow volume adjusting unit **100** adjusts a flow volume of warm/cold water to be spouted.
[0076] The flow volume adjusting unit **100** includes a motor **72**. The motor **72** is a stepping motor, for example, a rotational position (namely, drive amount) of the motor **62** is controlled by the number of steps. In accordance with an operation with respect to the water amount adjusting button **42** of the remote controller **4**, the motor **72** drives so as to drive a flow adjusting valve **102** (see FIG. **6**), and the flow volume adjusting unit **100** adjusts a flow volume of warm/cold water.
[0077] The spout water switching unit **30** switches between spouting and stopping of warm/cold water to be discharged from the combination faucet unit **50**. In other words, the spout water switching unit **30** switches between spouting and stopping of warm/cold water from the water spout parts **2**. The spout water switching unit **30** changes a water spouting part of warm/cold water. The faucet device **1** causes the spout water switching unit **30** to execute switching between spouting and stopping of warm/cold water from the water spout part **2**, and further causes the flow volume adjusting unit **100** to adjust a flow volume of warm/cold water when warm/cold water is spouted.
[0078] The spout water switching unit **30** includes a plurality of electromagnetic valves **31** to **35**. Specifically, the spout water switching unit **30** includes the first electromagnetic valve **31**, the second electromagnetic valve **32**, the third electromagnetic valve **33**, the fourth electromagnetic valve **34**, and the fifth electromagnetic valve **35**.
[0079] Each of the first electromagnetic valve **31** to the fourth electromagnetic valve **34** is turned into “closed (OFF)” or “open (ON)” in accordance with an operation with respect to the switching button **43**.
[0080] In a case where the first electromagnetic valve **31** to the fourth electromagnetic valve **34** are “closed”, warm/cold water is not spouted from the faucet **21**, the hand shower **22**, the overhead shower **23**, or the warm pillar **24**. In a case where one of the first electromagnetic valve **31** to the fourth electromagnetic valve **34** is “open”, warm/cold water is spouted from one of the faucet **21**, the hand shower **22**, the overhead shower **23**, and the warm pillar **24**, which is corresponding to the “open” electromagnetic valve.
[0081] The first electromagnetic valve **31** switches between spouting and stopping of warm/cold water from the faucet **21**. The second electromagnetic valve **32** switches between spouting and stopping of warm/cold water from the hand shower **22**. The third electromagnetic valve **33** switches between spouting and stopping of warm/cold water from the overhead shower **23**. The fourth electromagnetic valve **34** switches between spouting and stopping of warm/cold water from the warm pillar **24**.
[0082] For example, in a case where the first electromagnetic valve **31** is “open” and the second

electromagnetic valve **32** to the fourth electromagnetic valve **34** are "closed", warm/cold water is spouted from the faucet **21**.
[0083] The fifth electromagnetic valve **35** is switched between "closed (OFF)" and "open (ON)" in accordance with a change in a mode for spouting water by the remote controller **4** or an operation by an external device **6** (for example, remote controller arranged in water closet). The fifth electromagnetic valve **35** is a valve for discharging water remaining in a hose of the hand shower **22** or the like and/or a pipe, and further is usually kept "closed". In other words, the fifth electromagnetic valve **35** is turned into "open" in processing remaining water. In a case where the fifth electromagnetic valve **35** is turned into "open", remaining water is discharged from the hot water waiting water spout port **25***a* via the remaining water discharging flow path **25**.
[0084] In accordance with an operation of the faucet body **3** which is received by an operation of the remote controller **4**, the controller **7** controls the motors **62** and **72**, and the first electromagnetic valve **31** to the fourth electromagnetic valve **34**. In accordance with an operation of the remote controller **4** and/or the external device **6**, for example, the controller **7** controls the fifth electromagnetic valve **35**.
[0085] The controller **7** is a controller. The controller **7** includes a microcomputer including a Central Processing Unit (CPU), a Read Only Memory (ROM), a Random Access Memory (RAM), and the like; and various circuits, for example. Note that the controller **7** may include hardware such as an Application Specific Integrated Circuit (ASIC) and a Field Programmable Gate Array (FPGA).
[0086] The communication unit **5** receives an operation signal from the remote controller **4** and the external device **6**, and further transmits the received operation signal to the controller **7**. The remote controller **4** and the external device **6** are connected to the communication unit **5** by wired communication or wireless communication. The controller **7** is connected to the communication unit **5** by using wired communication or wireless communication.
[0087] Next, the faucet body **3** will be explained with reference to FIG. **4**. FIG. **4** is a perspective view illustrating the faucet body **3**. The faucet body **3** includes a flow path unit **9** in addition to the combination faucet unit **50**, the spout water switching unit **30**, and the controller **7** (see FIG. **3**). The faucet body **3** is attached to the wall part **14***a* (see FIG. **1**) of the bath unit **10** by the flow path unit **9**.
[0088] The flow path unit **9** includes the hot water supplying path **39** (see FIG. **3**) that causes hot water supplied from the hot water supplying source **37** (see FIG. **3**) to flow into the combination faucet unit **50**, the water supplying path **46** (see FIG. **3**) that causes cold water supplied from the water supply source **44** (see FIG. **3**) to flow into the combination faucet unit **50**, and a flow path that causes warm/cold water to flow into the water spout parts **2** from the combination faucet unit **50**. Note that the flow path unit **9** is provided with the spout water switching unit **30**. The flow path unit **9** is provided with the water stopping valves **38** and **45** (see FIG. **3**). The flow path unit **9** is provided with the remaining water discharging flow path **25**.
[0089] In the faucet body **3**, the flow path unit **9** is arranged in a back portion thereof, and the combination faucet unit **50** is arranged in front of the flow path unit **9**. A part of the flow path unit **9** is arranged above the combination faucet unit **50**. A control box **7***a* housing therein the controller **7** is arranged in front of the combination faucet unit **50**.
[0090] Next, with reference to FIG. **5** and FIG. **6**, the combination faucet unit **50** will be more specifically explained. FIG. **5** is a perspective view illustrating the combination faucet unit **50**. FIG. **6** is a perspective cross sectional view taken along a line VI-VI illustrated in FIG. **5**.
[0091] As illustrated in FIG. **5** and FIG. **6**, the combination faucet unit **50** includes a unit body **51**, a temperature adjusting side motor unit **60**, a flow adjusting side motor unit **70**, the hot water and cold water mixing unit **80**, and the flow volume adjusting unit **100**. In the combination faucet unit **50**, the unit body **51** is provided with a water spout port **52** so as to spout warm/cold water from the water spout port **52**.

[0092] The unit body **51** extends in the left-right direction. The hot water and cold water mixing unit **80** and the flow volume adjusting unit **100** are inserted into the unit body **51** in the left-right direction. In other words, insertion directions of the hot water and cold water mixing unit **80** and the flow volume adjusting unit **100** coincide with the left-right direction.
[0093] A hot water supplying port **53** connected to the hot water supplying path **39** (see FIG. **3**) is arranged on the left side of the unit body **51**. A cold water supplying port **54** connected to the water supplying path **46** (see FIG. **3**) is arranged in an intermediate portion in the left-right direction of the unit body **51**. In the unit body **51**, the hot water supplying port **53** and the cold water supplying port **54** are adjacently formed in the left-right direction.
[0094] The temperature adjusting side motor unit **60** includes a cover **61** and the motor **62**, and further is arranged in a left edge of the unit body **51**. The cover **61** is arranged in a left edge of the unit body **51** via a spacer **55**. The motor **62** is arranged in the cover **61**. The flow adjusting side motor unit **70** includes a cover **71** and the motor **72**, and further is arranged in a right edge of the unit body **51**. The cover **71** is arranged in a right edge of the unit body **51** via a spacer **56**. The motor **72** is arranged in the cover **71**.
[0095] The motor **62** is an actuator that causes the hot water and cold water mixing unit **80** to operate. The motor **62** switches between whether or not mixing cold water and hot water in accordance with a rotational position (namely, drive amount) of the motor **62**. In a case where spouting mixed hot water, the motor **62** sets a temperature of mixed hot water in accordance with a rotational position (namely, drive amount) of the motor **62**. The motor **72** adjusts a flow volume of warm/cold water in accordance with a rotational position (namely, drive amount) of the motor **72**.
[0096] The hot water and cold water mixing unit **80** includes a body casing **81**, the temperature adjusting valve **82**, a thermo-sensitive spring **83** (namely, thermo-sensitive energizing part), a bias spring **84**, a liner **85**, and a spindle **86**.
[0097] The body casing **81** includes a first body casing **87** and a second body casing **88**. A hot water inlet **87***a* that is a hot water inlet with respect to an internal space of the body casing **81**, and a cold water inlet **87***b* that is a cold water inlet with respect to the above-mentioned internal space are formed on a peripheral wall of the first body casing **87**. A mixed hot water and cold water outlet **88***a* is formed in a right end part of the second body casing **88**. The hot water inlet **87***a*, the cold water inlet **87***b*, and the mixed hot water and cold water outlet **88***a* are hole parts via which the internal space of the body casing **81** is communicated with the outside. In an intermediate portion of the body casing **81** in the left-right direction, the hot water inlet **87***a* and the cold water inlet **87***b* are formed such that the hot water inlet **87***a* is located on the left side of the cold water inlet **87***b*.
[0098] An outer peripheral space of the body casing **81** is partitioned by O-rings **89***a*, **89***b*, and **89***c* that are sealing members. Thus, a hot water dedicated circular flow path **89** facing the hot water inlet **87***a* and a cold water dedicated circular flow path **90** facing the cold water inlet **87***b* are formed in the outer peripheral space of the body casing **81**.
[0099] A spout water flow path **91** is formed in an internal space of the body casing **81**. The spout water flow path **91** can be communicated with the water spout port **52** via the mixed hot water and cold water outlet **88***a* and the flow adjusting valve **102**.
[0100] The thermo-sensitive spring **83** is housed in the second body casing **88**. The thermo-sensitive spring **83** is arranged in the spout water flow path **91**. The thermo-sensitive spring **83** is a spring whose spring constant changes in accordance with the temperature, for example, and further is made of Shape Memory Alloy (SMA). The thermo-sensitive spring **83** energizes the temperature adjusting valve **82** towards the left side.
[0101] The bias spring **84** is housed in the first body casing **87** arranged in the left portion of the second body casing **88**. The bias spring **84** is arranged in the spout water flow path **91**. The bias spring **84** is a spring whose spring constant is substantially constant with respect to the temperature. The bias spring **84** energizes the temperature adjusting valve **82** towards the right side.
[0102] The temperature adjusting valve **82** is arranged in the first body casing **87** on the right side

thereof. The temperature adjusting valve **82** is incorporated into the first body casing **87** to be able to slide along an axial direction (namely, left-right direction) thereof. The temperature adjusting valve **82** moves along the left-right direction in accordance with an energizing force of the bias spring **84** and an energizing force of the thermo-sensitive spring **83** so as to adjust a communication state between the hot water dedicated circular flow path **89** and the cold water dedicated circular flow path **90**, and the spout water flow path **91**.

[0103] Specifically, the temperature adjusting valve **82** is held at a position where an energizing force of the bias spring **84** and an energizing force of the thermo-sensitive spring **83** balance with each other. An open degree of the hot water inlet **87***a* is larger and further an open degree of the cold water inlet **87***b* is smaller as the temperature adjusting valve **82** further moves towards the right side, and thus an amount of hot water supplied to the spout water flow path **91** increases so as to reduce an amount of cold water, and thus a temperature of mixed hot water rises. An open degree of the hot water inlet **87***a* is smaller and further an open degree of the cold water inlet **87***b* is larger as the temperature adjusting valve **82** further moves towards the left side, and thus an amount of hot water supplied into the body casing **81** reduces so as to increase an amount cold water, and thus a temperature of mixed hot water reduces.

[0104] The liner **85** is in contact with an edge of the bias spring **84** on a side opposite to the temperature adjusting valve **82**, and further is connected with the motor **62** via the spindle **86**. The spindle **86** converts rotational movement of the motor **62** into linear movement of the liner **85** in the left-right direction. Thus, the liner **85** moves in the left-right direction in accordance with rotation of the motor **62**.

[0105] In the hot water and cold water mixing unit **80**, the liner **85** moves in the left-right direction in accordance with a rotational position of the motor **62**, so as to change a position of a left edge of the bias spring **84**. Thus, in accordance with a rotational position of the motor **62**, the hot water and cold water mixing unit **80** is capable of changing a position of the temperature adjusting valve **82** where an energizing force of the bias spring **84** and an energizing force of the thermo-sensitive spring **83** balance with each other. Therefore, in a case where spouting mixed hot water, the hot water and cold water mixing unit **80** is capable of setting a temperature of the mixed hot water to a temperature according to a rotational position of the motor **62**.

[0106] For example, in a case where a temperature of hot water is changed to change a temperature of mixed hot water in spouting the mixed hot water, the thermo-sensitive spring **83** expands or contracts in accordance with the temperature of mixed hot water, and the temperature adjusting valve **82** moves in the left-right direction so as to automatically change the balanced position of the temperature adjusting valve **82**. Therefore, amounts of hot water and cold water flowing into the spout water flow path **91** are adjusted so as to automatically adjust a temperature of mixed hot water.

[0107] The flow volume adjusting unit **100** includes a spindle **101** and the flow adjusting valve **102**. One edge of the spindle **101** is connected with the motor **72**, and the other edge thereof is connected with the flow adjusting valve **102**.

[0108] The flow adjusting valve **102** is arranged in the spout water flow path **91**. The flow adjusting valve **102** is rotated in accordance with rotation of the motor **72**. A communication opening **102***a* is formed in the flow adjusting valve **102**. The flow adjusting valve **102** is arranged so as to face the water spout port **52** of the unit body **51**. The flow adjusting valve **102** is rotated in accordance with rotation of the motor **72**, and further changes a communication area between the communication opening **102***a* and the water spout port **52** so as to change a flow volume.

[0109] Specifically, in a case where the motor **72** locates in a predetermined water stopping position, the communication opening **102***a* is not communicated with the water spout port **52**. Thus, in a case where the motor **72** locates in the predetermined water stopping position, warm/cold water is not spouted from the water spout part **2**.

[0110] In a case where the motor **72** rotates towards a water spouting position from a water stopping position, for example, the communication opening **102***a* becomes communicated with the water spout port **52**. Thus, warm/cold water is spouted from the water spout port **52**.

[0111] The flow adjusting valve **102** is capable of changing an area in which the communication opening **102***a* and the water spout port **52** are communicated with each other, in accordance with a rotational position of the motor **72**. In other words, the flow volume adjusting unit **100** is capable of adjusting a flow volume of warm/cold water to be spouted from the water spout parts **2**, in accordance with a rotational position of the motor **72**.

[0112] Note that in temperature and flow adjusting control of the faucet device **1**, the controller **7** does not execute a fine adjustment by using feedback control. In a case where a water spouting state is changed into a water stopping state, a process for returning to original positions (namely, positions to be control reference) of the motors **62** and **72** for each time of stopping water is not executed, step-out or the like occurs to be gradually accumulated, and thus open degrees of positions of the motors **62** and **72** are supposed to be shifted. In order to avoid the above-mentioned case, the controller **7** executes "origin searching" that is a process for returning positions of the motor **62** and the motor **72** to original positions thereof at predetermined time intervals, during calibration, or the like.

[0113] Next, temperature controlling of the faucet device **1** will be explained with reference to FIG. **7** to FIG. **13**. FIG. **7** is a flowchart illustrating a processing procedure for temperature controlling in a water spouting mode. FIG. **8** is a flowchart illustrating a processing procedure for an initial process and a supplied hot water temperature determining process in a calibration mode. FIG. **9** is a flowchart illustrating a processing procedure for a normal optimizing process in the calibration mode. FIG. **10** is a flowchart illustrating a processing procedure for a temperature adjustment table assigning process in the calibration mode. FIG. **11** to FIG. **13** are diagrams illustrating calculation of assignment factors.

[0114] The controller **7** is capable of executing a "water spouting mode" and a "calibration mode". The "water spouting mode" is a mode for driving, in a case where receiving information on a set temperature from the remote controller **4**, the motor **62** to a predetermined rotational position or by a predetermined drive amount alone, which is corresponding to the set temperature so as to adjust a position of the temperature adjusting valve **82** in an axial direction thereof. The "calibration mode" is a mode for adjusting setting of a predetermined rotational position or a predetermined drive amount in accordance with at least one of an arrangement environment (for example, water pressure, hot water supplying temperature) and product unevenness.

[0115] A processing procedure for temperature adjustment in the "water spouting mode" will be first explained.

[0116] As illustrated in FIG. **7**, the controller **7** receives a request for changing temperature adjustment setting from the remote controller **4** (Step S**101**).

[0117] Next, the controller **7** acquires a temperature adjustment setting value from the remote controller **4** (Step S**102**). The temperature adjustment setting value means a value of a set temperature which is transmitted from the remote controller **4**.

[0118] Next, the controller **7** acquires a temperature adjustment correcting value from the remote controller **4** (Step S**103**). The temperature adjustment correcting value is a value for correcting a temperature adjustment setting value into a higher temperature or a lower temperature. The temperature adjustment correcting value is preliminarily set by the remote controller **4**.

[0119] Next, the controller **7** drives the motor **62** into a predetermined rotational position on the basis of a temperature adjusting open degree table (Step S**104**), and further ends the processing. The temperature adjusting open degree table means data that stores therein set temperatures and rotational positions of the motor **62** in association with each other, for example. For example, in a case where a temperature adjustment setting value is 40° C. and a temperature adjustment correcting value is +1° C., the controller **7** refers to a table value (namely, rotational position) at 41° C. in the temperature adjusting open degree table.

[0120] According to such temperature controlling, the faucet device **1** does not have to always recognize the present output, unlike feedback control. Moreover, the faucet device **1** does not loop the processing, so that it is possible to execute implementation by simple processing.
[0121] Next, the "calibration mode" will be specifically explained. The "calibration mode" is executed by an operation by a user with respect to the remote controller **4**. Specifically, in a case where a user simultaneously holds down a low temperature button **41***b* and the water increasing button **42***a* of the remote controller **4** for three seconds so as to shift to a set screen, and then selects "calibration" from the set screen; a display (not illustrated) displays thereon execution necessity of the "calibration mode". In a case where a user selects "execute calibration mode" in the remote controller **4**, the calibration mode is executed. During execution of the calibration mode, the remote controller **4** causes the display (not illustrated) to display "executing calibration mode". Calibration can be executed by the remote controller **4** that is used in common water spouting operations, and thus the faucet device **1** does not need a dedicated setting remote controller. Note that execution of the calibration mode may be temporarily interrupted or cancelled during execution of the calibration mode by an operation of a user with respect to the remote controller **4**.
[0122] In the "calibration mode", an initial process (Step S**201** to Step S**204**) and a supplied hot water temperature determining process (Step S**205** to Step S**209**) are first executed.
[0123] As illustrated in FIG. **8**, the controller **7** executes origin searching on the motors **62** and **72**, and further opens the fifth electromagnetic valve **35** (Step S**201**). The faucet device **1** opens the fifth electromagnetic valve **35** so as to execute calibration while spouting water from the hot water waiting water spout port **25***a*. The controller **7** drives the motor **62** while recognizing environment in actually spouting water, so that it is possible to improve the accuracy in the calibration.
[0124] Next, the controller **7** drives the motor **62** so as to obtain a first predetermined temperature (Step S**202**). For example, the first predetermined temperature is set to 42° C.
[0125] Next, the controller **7** determines whether or not the first predetermined time interval has elapsed since the motor **62** is driven (Step S**203**). For example, the first predetermined time interval is set to five minutes.
[0126] In a case where determining that the first predetermined time interval has elapsed since the motor **62** is driven (Step S**203**: Yes), the controller **7** ends the calibration process. In this case, the controller **7** ends the processing without updating the temperature adjusting open degree table.
[0127] In a case where determining that the first predetermined time interval has not elapsed since the motor **62** is driven (Step S**203**: No), the controller **7** determines whether or not a temperature (namely, hot water temperature) of hot water detected by the temperature sensor **48** is higher than a second predetermined temperature (Step S**204**). For example, the second predetermined temperature is set to 32° C. The determination in Steps S**203** and S**204** is objected to determination of whether or not a power source of a hot water supplying device, which is the hot water supplying source **37**, is turned ON.
[0128] In a case where determining that a temperature of hot water detected by the temperature sensor **48** is equal to or less than the second predetermined temperature (Step S**204**: No), the controller **7** returns the processing to Step S**203**.
[0129] In a case where determining that a temperature of hot water detected by the temperature sensor **48** is higher than the second predetermined temperature (Step S**204**: Yes), the controller **7** determines whether or not a hot water supplying temperature is stable at equal to or more than a third predetermined temperature (Step S**205**). For example, the controller **7** is capable of determining whether or not a hot water supplying temperature is stable, on the basis of the fact that a temperature of hot water detected by the temperature sensor **48** falls within a predetermined range for a predetermined time interval.
[0130] In a case where determining that a hot water supplying temperature is stable at equal to or more than the third predetermined temperature (Step S**205**: Yes), the controller **7** executes origin searching on the motor **62** (Step S**206**), and further shifts the processing to a normal optimizing

process illustrated in FIG. **9**. As described above, after the hot water supplying temperature becomes stable, the controller **7** starts to adjust setting of a rotational position of the motor **62**. The faucet device **1** is capable of precisely executing temperature controlling in calibration while reducing effects due to heat absorption of cold cast metal in a case where the flow path unit **9** (see FIG. **4**) is constituted of cast metal, and starting up of the hot water supplying device. Thus, the controller **7** is capable of improving the accuracy in calibration.

[0131] In a case where determining that a hot water supplying temperature is not stabilized at equal to or more than the third predetermined temperature (Step S**205**: No), the controller **7** determines that the hot water supplying temperature is equal to or less than an appropriate temperature water supply, and further assigns a fixed table to a temperature adjusting open degree table (Step S**207**). The fixed table is preliminarily set on the basis of evaluation data under predetermined conditions (for example, water temperature of water supplying path **46** is 15° C. and hot water temperature of hot water supplying path **39** is 40° C., which are under pressure of 0.2 MPa). As described above, in a case where a hot water supplying temperature is lower than the third predetermined temperature, the controller **7** executes controlling with the use of a dedicated table. Thus, the faucet device **1** is capable of recognizing whether or not a hot water supplying temperature is an appropriate temperature, and further is capable of causing the remote controller **4** to display whether or not the hot water supplying temperature is an appropriate temperature in accordance with a situation, so as to report it to a user. In a case where a hot water supplying temperature is lower than the third predetermined temperature, the faucet device **1** is capable of reducing a case where lukewarm hot water is spouted. Thus, the faucet device **1** is capable of avoiding reduction in usability of a user.

[0132] Next, the controller **7** executes origin searching on the motor **62** (Step S**208**).

[0133] Next, the controller **7** drives the motors **62** and **72** to default positions thereof (Step S**209**), and further ends the calibration process. For example, the default position of the motor **62** is a rotational position of the motor **62** corresponding to a center value (for example, 40° C.) of set temperature. The default position of the motor **72** is a rotational position of the motor **72** corresponding to a center value of flow volume.

[0134] Next, a processing procedure for a normal optimizing process in the “calibration mode” will be explained.

[0135] As illustrated in FIG. **9**, the controller **7** executes PID control with respect to a first target temperature (Step S**210**). Specifically, the controller **7** calculates, by using PID control, a rotational position of the motor **62** whose target temperature is the first target temperature, and further drives the motor **62** to the calculated rotational position. For example, the first target temperature is set to 35° C. Note that PID control is employed as control for coinciding a spouted water temperature with a target temperature; however, not limited thereto, a spouted water temperature may be adjusted into the target temperature by various control methods.

[0136] Next, the controller **7** determines whether or not PID control has continued for a third predetermined time interval since the PID control of Step S**210** is started (Step S**211**). For example, the third predetermined time interval is set to 60 seconds.

[0137] In a case where determining that the third predetermined time interval has not continued since PID control of Step S**210** is started (Step S**211**: No), the controller **7** determines whether or not a deviation e in the PID control is equal to or less than a fourth predetermined temperature, and the state (hereinafter, may be simply referred to as “state equal to or less than fourth predetermined temperature”) has continued for a fourth predetermined time interval (Step S**212**).

[0138] In a case where determining that a state equal to or less than the fourth predetermined temperature has not continued for the fourth predetermined time interval (Step S**212**: No), the controller **7** returns the processing to Step S**211**.

[0139] Next, in a case where determining that the third predetermined time interval has continued since PID control is started with respect to the first target temperature in Step S**210** (Step S**211**:

Yes), or in a case where determining that a state equal to or less than the fourth predetermined temperature has continued for the fourth predetermined time interval (Step S**212**: Yes); the controller **7** records data on the basis of search result (Step S**213**). Specifically, the controller **7** records, as a temperature changing point, a drive amount (namely, the number of steps) of the motor **62** and a temperature (namely, first temperature) of mixed hot water detected by the temperature sensor **49** at a timing of Step S**213**.

[0140] Next, the controller **7** executes PID control with respect to a second target temperature (Step S**214**). Specifically, the controller **7** calculates, by using PID control, a rotational position of the motor **62** whose target temperature is the second target temperature, and further drives the motor **62** to the calculated rotational position. For example, the second target temperature is set to 40° C.

[0141] Next, the controller **7** determines whether or not the third predetermined time interval has continued since PID control in Step S**214** is started (Step S**215**).

[0142] In a case where determining that the third predetermined time interval has not continued since PID control in Step S**214** is started (Step S**215**: No), the controller **7** determines whether or not a state equal to or less than the fourth predetermined temperature has continued for the fourth predetermined time interval (Step S**216**).

[0143] In a case where determining that a state equal to or less than the fourth predetermined temperature has not continued for the fourth predetermined time interval (Step S**216**: No), the controller **7** returns the processing to Step S**215**.

[0144] In a case where determining that the third predetermined time interval has continued since PID control in Step S**214** is started (Step S**215**: Yes), or in a case where determining that a state equal to or less than the fourth predetermined temperature has continued for the fourth predetermined time interval (Step S**216**: Yes), the controller **7** records data on the basis of search result (Step S**217**). Specifically, the controller **7** records, as a temperature changing point, a drive amount (namely, the number of steps) of the motor **62** and a temperature (namely, second temperature) of mixed hot water detected by the temperature sensor **49** at a timing of Step S**217**.

[0145] Next, the controller **7** executes PID control with respect to a third target temperature (Step S**218**). Specifically, the controller **7** calculates, by using PID control, a rotational position of the motor **62** whose target temperature is the third target temperature, and further drives the motor **62** to the calculated rotational position. For example, the third target temperature is set to 45° C.

[0146] Next, the controller **7** determines whether or not the third predetermined time interval has continued since PID control in Step S**218** is started (Step S**219**).

[0147] In a case where determining that the third predetermined time interval has continued since PID control in Step S**218** is started (Step S**219**: No), the controller **7** determines whether or not a rotational position of the motor **62** reaches an upper limit rotational position (Step S**220**).

[0148] In a case where determining that a rotational position of the motor **62** does not reach the upper limit rotational position (Step S**220**: No), the controller **7** determines whether or not a state equal to or less than the fourth predetermined temperature has continued for the fourth predetermined time interval (Step S**221**).

[0149] In a case where determining that a state equal to or less than the fourth predetermined temperature has not continued for the fourth predetermined time interval (Step S**221**: No), the controller **7** returns the processing to Step S**219**.

[0150] In a case where determining that the third predetermined time interval has continued since PID control in Step S**218** is started (Step S**219**: Yes), or in a case where a state equal to or less than the fourth predetermined temperature has continued for the fourth predetermined time interval (Step S**221**: Yes), the controller **7** records data on the basis of search result (Step S**222**), and further proceeds to a normal optimizing process illustrated in FIG. **10**. Specifically, in Step S**222**, the controller **7** records a drive amount (namely, the number of steps) of the motor **62** and a temperature (namely, third temperature) of mixed hot water detected by the temperature sensor **49** at a timing of Step S**222**.

[0151] In a case where determining reaching an upper limit rotational position (Step S**220**: Yes), the controller **7** waits for a fifth predetermined time interval (Step S**223**). For example, the fifth predetermined time interval is set to 30 seconds.
[0152] Next, the controller **7** sets an upper limit rotational position to a rotational position at the third target temperature (Step S**224**).
[0153] Next, the controller **7** records data on the basis of search result (Step S**222**), and further proceeds to the normal optimizing process illustrated in FIG. **10**. Specifically, in Step S**222**, the controller **7** records, as a temperature changing point, an upper limit rotational position of the motor **62** and a temperature (namely, third temperature) of mixed hot water detected by the temperature sensor **49**.
[0154] As illustrated in FIG. **11**, temperature changing points are plotted on a graph whose lateral axis indicates a rotational position of the motor **62** and whose vertical axis indicates a temperature of mixed hot water detected by the temperature sensor **49** while indicating a temperature changing point recorded in Step S**213** as a point A, indicating a temperature changing point recorded in Step S**217** as a point B, and indicating a temperature changing point recorded in Step S**222** as a point C.
[0155] Next, a processing procedure for a temperature adjustment table assigning process in the "calibration mode" will be explained.
[0156] As illustrated in FIG. **10**, the controller **7** calculates assignment factors (Step S**225**). Specifically, the controller **7** calculates a slope and an intercept of a linear equation of a straight line AB obtained by connecting the point A (s**1**, t**1**) and the point B (s**2**, t**2**) illustrated in FIG. **12**, and further calculates a slope and an intercept of a linear equation of a straight line BC obtained by the point B (s**2**, t**2**) and the point C (s**3**, t**3**) illustrated in FIG. **13**. The assignment factors include a slope and an intercept of a linear equation of the straight line AB, and a slope and an intercept of a linear equation of the straight line BC.
[0157] Next, the controller **7** closes the fifth electromagnetic valve **35** (Step S**226**). The controller **7** closes the fifth electromagnetic valve **35** so as to stop spouting water from the hot water waiting water spout port **25***a*.
[0158] Next, the controller **7** interpolate table values (Step S**227**). Specifically, on the basis of an assignment factor calculated in Step S**225**, the controller **7** calculates a rotational position of the motor **62** for each set temperature (namely, for each temperature to be set by remote controller **4**). For example, at a set temperature equal to or less than the second target temperature, the controller **7** calculates a rotational position of the motor **62** on the basis of a linear equation of the straight line AB. At a set temperature higher than the second target temperature, the controller **7** calculates a rotational position of the motor **62** on the basis of a linear equation of the straight line BC.
[0159] Note that a rotational position of the motor **62** is calculated on the basis of three temperature changing points; however, not limited thereto, equal to or more than three temperature changing points may be searched, a linear equation of a straight line between each two of the temperature changing points in descending order of a target temperature may be calculated, and a rotational position of the motor **62** may be calculated for each set temperature similarly to the case of three points. The controller **7** uses equal to or more than three temperature changing points so as to improve the accuracy in calibration.
[0160] Next, the controller **7** updates whole of the table data (Step S**228**). Specifically, the controller **7** updates whole of the data on the temperature adjusting open degree table by using a rotational position of the motor **62** calculated in Step S**227**.
[0161] Next, the controller **7** executes origin searching on the motor **62** (Step S**229**).
[0162] Next, the controller **7** drives the motors **62** and **72** to default positions thereof (Step S**230**), and further ends the processing.
[0163] As described above, in the faucet device **1**, in a case where receiving information on a set temperature, the controller **7** drives the motor **62** to a predetermined rotational position or by a predetermined drive amount alone, which is corresponding to the set temperature, so as to execute

the water spouting mode for adjusting a position of the temperature adjusting valve **82** in an axial direction thereof, and the calibration mode for adjusting setting of a predetermined position or a predetermined drive amount in accordance with at least one of an arrangement environment and product unevenness.

[0164] By executing control in such a manner, the faucet device **1** is capable of adjusting, in accordance with each of scenes, a hot water temperature, a hot water pressure, a cold water temperature, and a cold water pressure from the hot water supplying path **39** and the water supplying path **46** which are different for each scene, and individual difference in temperature adjustment of an SMA thermo-valve. Thus, the faucet device **1** causes the motor **62** to move the temperature adjusting valve **82** only when changing a set temperature, and further executes temperature adjustment by using the thermo-sensitive spring **83** during spouting of water, so that it is possible to execute temperature adjustment having high accuracy according to the set temperature. The faucet device **1** is capable of achieving, as much as possible, features to be capable of shortening a time interval until a temperature becomes stable in a case where temperature change and/or pressure fluctuation occurs in supplied cold water or supplied hot water during spouting of water.

[0165] The controller **7** includes the temperature sensors **48** and **49** that are respectively arranged on upstream and downstream of the hot water and cold water mixing unit **80**, and in the calibration mode, setting of a predetermined rotational position or a predetermined drive amount is adjusted on the basis of information transmitted from the temperature sensor **48**.

[0166] By executing control in such a manner, in the faucet device **1**, in entering the calibration mode, a predetermined position or a predetermined drive amount is automatically adjusted. Thus, the faucet device **1** is capable of improving usability of a user.

[0167] In the calibration mode, the controller **7** acquires a first temperature that is a temperature at a temperature changing point where searching and specifying a temperature lower than the second target temperature as a target temperature, a second temperature that is a temperature at a temperature changing point where searching and specifying the second target temperature as a target temperature, and a third temperature that is a temperature at a temperature changing point where searching and specifying a temperature higher than the second target temperature as a target temperature, so as to adjust setting of a predetermined rotational position or a predetermined drive amount on the basis of the second and first temperatures and the second and third temperatures.

[0168] By executing control in such a manner, it is true that there presents a case where characteristics of the thermo-sensitive spring **83** change between under and over the second target temperature (for example, 40° C.), the faucet device **1** adjusts setting of a rotational position or a drive amount that are predetermined on the basis of a temperature (namely, temperature on straight line AB) within a range lower than the second target temperature and a temperature (namely, temperature on straight line BC) within a range higher than the second target temperature, so that it is possible to prevent a set temperature from shifting in executing temperature adjustment crossing a predetermined temperature.

[0169] Note that the actuator is the motor **62**, in the calibration mode, the temperature sensor **49** arranged on downstream of the hot water and cold water mixing unit **80** detects a spouted water temperature, and the controller **7** adjusts setting of a predetermined rotational position or a predetermined drive amount on the basis of a drive amount (namely, the number of steps) of the motor **62** and a spouted water temperature when the motor **62** takes the above-mentioned drive amount (namely, the number of steps).

[0170] By executing control in such a manner, the faucet device **1** executes simple operational processing on the basis of a drive amount (namely, the number of steps) of the motor **62** and a spouted water temperature, so that it is possible to simplify software processing.

[0171] During the calibration mode, the controller **7** does not release the excitation of the motor **62**. In other words, the controller **7** keeps an energized state of the motor **62** during the calibration

mode.
[0172] By executing control in such a manner, in a case where the excitation is released after driving the motor **62** by a small step, there presents possibility in the faucet device **1** that spring back occurs, which causes the motor **62** to return to the last position due to load that is applied to the motor **62**; however, it is possible to remove effects due to the above-mentioned spring back, and further to execute precise temperature controlling during execution of calibration. Thus, the faucet device **1** is capable of improving the accuracy in calibration.
[0173] The number of temperature changing points in the above-mentioned calibration mode is three; however, may be two or one.
[0174] Specifically, in the calibration mode, the controller **7** may search two temperature changing points, and further may adjust setting of a predetermined rotational position or a predetermined drive amount on the basis of the specified temperature changing points. For example, the two temperature changing points may be temperatures near the second target temperature (for example, 40° C.). Similar to the calibration in a case where the number of temperature changing points is three, the controller **7** calculates an assignment factor of two temperature changing points so as to calculate a rotational position of the motor **62** for each set temperature.
[0175] By executing control in such a manner, the faucet device **1** is capable of executing calibration without taking time, while keeping the accuracy in calibration near the second target temperature.
[0176] Alternatively, in the calibration mode, the controller **7** may search a single temperature changing point, and further may adjust setting of a predetermined rotational position or a predetermined drive amount on the basis of the specified temperature changing point. For example, the single temperature changing point may be a temperature whose target temperature is the second target temperature (for example, 40° C.). The controller **7** calculates a rotational position of the motor **62** for each set temperature, on the basis of a single temperature changing point and a preliminarily-set assignment factor.
[0177] By executing control in such a manner, the faucet device **1** is capable of shortening a time interval until completion of calibration.
[0178] According to one aspect of the embodiments, it is possible to shorten a time interval until a spouted water temperature stabilizes, and further to improve the accuracy in temperature adjustment.
[0179] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.
<Supplementary Note>
(1)
[0180] A faucet device including: [0181] a water spout part that spouts water towards an interior of a bathroom; [0182] a hot water and cold water mixing unit that mixes warm water and cold water to be supplied to the water spout part; [0183] an actuator that causes the hot water and cold water mixing unit to operate; [0184] a controller configured to control driving of the actuator; and [0185] an operation unit configured to transmit information on a set temperature to the controller based on an operation of a user, wherein [0186] the hot water and cold water mixing unit includes: [0187] a body casing in which a hot water inlet, a cold water inlet, and a mixed hot water and cold water outlet are formed; [0188] a thermo-sensitive energizing part that changes in accordance with a temperature of mixed hot water to be capable of adjusting open degrees of the hot water inlet and the cold water inlet; and [0189] a valve body that is incorporated into the body casing to be able to slide along an axial direction of the body casing to adjust the open degrees of the hot water inlet and the cold water inlet, and [0190] the controller is further configured to: [0191] execute a water spouting mode and a calibration mode, the water spouting mode being a mode for driving, in

receiving the information on the set temperature, the actuator to a predetermined position or by a predetermined drive amount that are corresponding to the set temperature to adjust a position of the valve body in an axial direction of the valve body; and the calibration mode is a mode for adjusting setting of the predetermined position or the predetermined drive amount in accordance with at least one of an arrangement environment and product unevenness.

(2)

[0192] The faucet device according to (1) further including: [0193] a temperature sensor arranged on upstream or downstream of the hot water and cold water mixing unit, wherein [0194] in the calibration mode, setting of the predetermined position or the predetermined drive amount is adjusted based on information from the temperature sensor.

(3)

[0195] The faucet device according to (1) or (2), wherein [0196] in the calibration mode, the controller is further configured to: [0197] cause the actuator to operate while spouting water from the water spout part.

(4)

[0198] The faucet device according to any one of (1) to (3), wherein [0199] the calibration mode can be executed by an operation with respect to the operation unit.

(5)

[0200] The faucet device according to any one of (1) to (4), wherein [0201] the controller is further configured to: [0202] in the calibration mode; [0203] search equal to or more than three temperature changing points; and [0204] adjust the predetermined position or the predetermined drive amount based on the specified temperature changing points.

(6)

[0205] The faucet device according to (5), wherein [0206] the controller is further configured to: [0207] in the calibration mode; [0208] acquire a first temperature that is a temperature at a temperature changing point where searching and specifying a temperature lower than a predetermined temperature as a target temperature; a second temperature that is a temperature at a temperature changing point where searching and specifying the predetermined temperature as a target temperature; and a third temperature that is a temperature at a temperature changing point where searching and specifying a temperature higher than the predetermined temperature as a target temperature; and [0209] adjust setting of the predetermined position or the predetermined drive amount based on the second and the first temperatures, and the second and the third temperatures.

(7)

[0210] The faucet device according to according to any one of (1) to (4), wherein [0211] the controller is further configured to: [0212] in the calibration mode; [0213] search two temperature changing points; and [0214] adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing points.

(8)

[0215] The faucet device according to any one of (1) to (4), wherein [0216] the controller is further configured to: [0217] in the calibration mode; [0218] search a single temperature changing point; and [0219] adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing point.

(9)

[0220] The faucet device according to any one of (1) to (8), wherein [0221] the actuator includes a motor, and [0222] in the calibration mode; [0223] a temperature sensor arranged on downstream of the hot water and cold water mixing unit detects a spouted water temperature, and [0224] the controller is further configured to: [0225] adjust setting of the predetermined position or the predetermined drive amount based on a drive amount of the motor and the spouted water temperature in a case where the motor takes the drive amount.

(10)
[0226] The faucet device according to any one of (1) to (3), wherein [0227] in the calibration mode; [0228] a temperature sensor arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and [0229] the controller is further configured to: [0230] determine whether or not the hot water supplying temperature is equal to or less than a predetermined temperature.
(11)
[0231] The faucet device according to (10), wherein [0232] the controller is further configured to: [0233] in the calibration mode; [0234] in a case where the hot water supplying temperature is equal to or less than the predetermined temperature, execute the water spouting mode by using a table that stores therein the set temperature, and the predetermined position or the predetermined drive amount in association with each other.
(12)
[0235] The faucet device according to any one of (1) to (11), wherein [0236] the actuator includes a motor, and [0237] the controller is further configured to: [0238] during the calibration mode, not release excitation of the motor.
(13)
[0239] The faucet device according to any one of (1) to (12), wherein [0240] in the calibration mode; [0241] a temperature sensor that is arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and [0242] the controller is further configured to: [0243] start to adjust setting of the predetermined position or the predetermined drive amount after the hot water supplying temperature becomes stable.

Claims

1. A faucet device comprising: a water spout part that spouts water towards an interior of a bathroom; a hot water and cold water mixing unit that mixes hot water and cold water to be supplied to the water spout part; an actuator that causes the hot water and cold water mixing unit to operate; a controller configured to control driving of the actuator; and an operation unit configured to transmit information on a set temperature to the controller based on an operation of a user, wherein the hot water and cold water mixing unit includes: a body casing in which a hot water inlet, a cold water inlet, and a mixed hot water and cold water outlet are formed; a thermo-sensitive energizing part that changes in accordance with a temperature of mixed hot water to be capable of adjusting open degrees of the hot water inlet and the cold water inlet; and a valve body that is incorporated into the body casing to be able to slide along an axial direction of the body casing to adjust the open degrees of the hot water inlet and the cold water inlet, and the controller is further configured to: execute a water spouting mode and a calibration mode, the water spouting mode being a mode for driving, in receiving the information on the set temperature, the actuator to a predetermined position or by a predetermined drive amount that are corresponding to the set temperature to adjust a position of the valve body in an axial direction of the valve body; and the calibration mode is a mode for adjusting setting of the predetermined position or the predetermined drive amount in accordance with at least one of an arrangement environment and product unevenness.
2. The faucet device according to claim 1 further comprising: a temperature sensor arranged on upstream or downstream of the hot water and cold water mixing unit, wherein in the calibration mode, setting of the predetermined position or the predetermined drive amount is adjusted based on information transmitted from the temperature sensor.
3. The faucet device according to claim 2, wherein in the calibration mode, the controller is further configured to: cause the actuator to operate while spouting water from the water spout part.
4. The faucet device according to claim 1, wherein the calibration mode can be executed by an

operation with respect to the operation unit.
5. The faucet device according to claim 3, wherein the controller is further configured to: in the calibration mode; search equal to or more than three temperature changing points; and adjust the predetermined position or the predetermined drive amount based on the specified temperature changing points.
6. The faucet device according to claim 5, wherein the controller is further configured to: in the calibration mode; acquire a first temperature that is a temperature at a temperature changing point where searching and specifying a temperature lower than a predetermined temperature as a target temperature; a second temperature that is a temperature at a temperature changing point where searching and specifying the predetermined temperature as a target temperature; and a third temperature that is a temperature at a temperature changing point where searching and specifying a temperature higher than the predetermined temperature as a target temperature; and adjust setting of the predetermined position or the predetermined drive amount based on the second and the first temperatures, and the second and the third temperatures.
7. The faucet device according to claim 3, wherein the controller is further configured to: in the calibration mode; search two temperature changing points; and adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing points.
8. The faucet device according to claim 3, wherein the controller is further configured to: in the calibration mode; search a single temperature changing point; and adjust setting of the predetermined position or the predetermined drive amount based on the specified temperature changing point.
9. The faucet device according to claim 3, wherein the actuator includes a motor, and in the calibration mode; a temperature sensor arranged on downstream of the hot water and cold water mixing unit detects a spouted water temperature, and the controller is further configured to: adjust setting of the predetermined position or the predetermined drive amount based on a drive amount of the motor and the spouted water temperature in a case where the motor takes the drive amount.
10. The faucet device according to claim 3, wherein in the calibration mode; a temperature sensor arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and the controller is further configured to: determine whether or not the hot water supplying temperature is equal to or less than a predetermined temperature.
11. The faucet device according to claim 10, wherein the controller is further configured to: in the calibration mode; in a case where the hot water supplying temperature is equal to or less than the predetermined temperature, execute the water spouting mode by using a table that stores therein the set temperature, and the predetermined position or the predetermined drive amount in association with each other.
12. The faucet device according to claim 3, wherein the actuator includes a motor, and the controller is further configured to: during the calibration mode, not release excitation of the motor.
13. The faucet device according to claim 3, wherein in the calibration mode; a temperature sensor that is arranged on upstream of the hot water and cold water mixing unit detects a hot water supplying temperature, and the controller is further configured to: start to adjust setting of the predetermined position or the predetermined drive amount after the hot water supplying temperature becomes stable.